

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-177***

“Kings Creek 25 kV”

December 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

CleanBay BioFuels, Inc. d/b/a CleanBay Renewables, the Interconnection Customer (IC), has proposed a 4.0 MW MFO (4.0 MWC) methane gas (biomass) fueled generating facility to be located in Westover, Maryland. AC1-177 is an upgrade request to the IC's prior queue project AA2-130 utilizing the same Point of Interconnection. The output of the combined queue projects will be a 6 MW Energy, 6 MW Capacity generating facility. PJM studied AC1-177 as a 4.0 MW injection into the Delmarva Power and Light Company's (DPL) system at the Kings Creek 25 kV Substation and evaluated it for compliance with reliability criteria for summer peak conditions in 2020. The planned in-service date, as requested by the IC, is June 24, 2018. This date is not be attainable due to additional required PJM studies and Transmission Owner construction schedules.

Point(s) of Interconnection

The Interconnection Customer requested a distribution level interconnection at the same Point of Interconnection prior queue project AA2-130. As a result, AC1-177 will interconnect with the Delmarva Power and Light distribution system as follows:

The first two MWs will connect to existing feeder MD2277, (along with prior queue project AA2-130) from the Kings Creek 25 kV T2 transformer; the next two MWs will connect to the Kings Creek 25 kV T2 transformer via a new express feeder (see Attachment 1).

Direct Connection Requirements

Criteria Limits for Distributed Energy Resource (DER) Connections to the ACE, DPL and Pepco Distribution Systems (less than 69kV)

1. Single Phase Limit

Any DER with a capacity that exceeds 100kW shall be a balanced 3 phase system.

2. Voltage Limits

DER's are permitted to cause a voltage fluctuation of up to 2% at the Point of Interconnection, ½ the band width of any voltage regulator at its terminals, and ½ the net dead band of a switched capacitor bank at its connection point. When a DER is at maximum output, it shall not raise the feeder voltage above the ANSI C84.1 or state limit, whichever is more conservative.

3. Existing Distribution Circuit Capacity Limits

The aggregate limit of large (250 kW and over) generators running in parallel with a single, existing distribution circuit is 0.5 MWs on the 4kV, 3MWs on the 12 kV, 6 MWs on the 25 kV, and 10 MWs on the 34 kV.

4. Express Circuit Capacity Limits

Distributed generation installations which exceed the limit for an existing circuit require an express circuit.

The maximum generator size for express circuits shall be:

- 4 kV 0.5 MW
- 12 – 13.8 kV 10 MW
- 23 – 25 kV 10 MW
- 33.26 – 34.5 kV 15 MW

5. Distribution Power Transformer Limit

The aggregate limit of large (250 kW and over) generator injection to a single distribution transformer of 22.5 MVA nameplate or larger is 10 MWs. Transformers with nameplate ratings lower than 22.5 MVA will be given lower ratings on an individual basis. If the transformer rating is significantly greater than 40 MVA it may be possible to consider a greater generation capacity.

Adding a new transformer will be considered if there is no availability on any of the existing transformers and space is available in an existing substation. Any proposed transformers would be PHI's standard distribution transformer.

6. Express Circuit Length Limit

If there is no space for an additional transformer at the closest substation, the next closest substation will be considered. The length of an express circuit is limited to 5 miles, or for the sake of the feasibility study, 3.8 straight line miles to the substation. This simplification is used because the feasibility study phase does not allow for the time and resources to examine routes in detail (including existing pole lines, easements, ROW, and environmental issues etc.)

7. When a New Substation is Required

If a distribution express circuit can't be built from an existing substation for a project, it will be necessary to construct a new distribution substation with a standard ring bus design. It will be supplied by extending existing transmission lines. It is the developer's responsibility to verify eligibility of this configuration for solar renewable energy certificates.

All limits, given above in MWs, are subject to more detailed study to ensure feasibility.

Transmission Owner Scope of Work

AC1-177 will utilize the same point of interconnection as prior queue project AA2-130 for the first two MWs.

The existing feeder MD2277 can only accommodate an additional 2 MWs of large generation. TO work required to accommodate 2 MW of generation on the existing feeder from Kings Creek Substation T2:

1. Protection, Planning, and other engineering departments will perform studies, design work, and prepare engineering estimates.
2. The operation of capacitors in this circuit needs to be reviewed in depth. Potential upgrades maybe required.

Estimated Costs	
Kings Creek Substation T2 MD2277	
Various Department Work	\$10,000
Capacitor Study & Upgrades	\$50,000
Subtotal Cost	\$60,000
Approximate Cost with 18% Overheads	\$70,800

The estimated time to complete this work is **24-36 months** after receipt of a fully executed interconnection agreement.

An express feeder is required to be built to accommodate the additional requested 2 MW of generation. TO work required to accommodate 2 MW of generation on an express feeder from Kings Creek Substation T2:

1. Design and construct one new 25 kV feeder with 477 aluminum conductor from Kings Creek Substation to the generation site – approximately 2 miles.
2. One new 25 kV feeder terminal position will be constructed.
3. A utility operated recloser equipped with the proper relaying and communications will be required.
4. Utility grade primary metering will be required.
5. Generation telemetry and remote trip capability will be provided to the control center.
6. A detailed, time-based study may be performed during later study phases.
7. Protection, Planning, and other engineering departments will perform studies, design work, and prepare engineering estimates.

8. Transfer trip will be required. The cost included below is to install 48SM ADSS fiber optic cable in conduit from Kings Creek Substation to the POI, a total distance of approximately 2 miles. 138 kV potential transformers will need to be installed if none already in-service.

Ball Park Costs			
KingsCreek Substation T2			
477 AAC Express Feeder	2	Miles	\$800,000
Fiber Installation			\$100,000
New Feeder Terminal			\$300,000
Substation Relaying & 138 kV PTs			\$200,000
Recloser & Metering			\$80,000
SCADA Integration into EMS			\$10,000
Dynamic Study			\$50,000
Various Departments Work			\$90,000
Subtotal Cost			\$1,630,000
Approximate Total Cost with 18% Contingency			\$1,874,500

The estimated time to complete this work is **36-48 months** after receipt of a fully executed interconnection agreement.

Interconnection Customer Scope of Work

The Interconnection Customer (IC) is responsible for all design and construction related to activities on their side of the point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC.

Protective relaying and metering design and installation must comply with PHI's applicable standards. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

Equipment Requirements

Any transformers on the IC's side must be Wye grounded on the utility side or alternatively 3 phase potential transformers and a relay capable of detecting over/under voltage shall be installed to detect an undesirable condition on the high side of the IC's transformer.

High Voltage Warning

Voltage received at the meter from the utility can be 104% or 105% of nominal. Normal operating procedures dictate that voltage at the substation be raised to the higher end of an acceptable bandwidth in order to provide adequate supply to distant customers. Transformers with no load taps should be used to reduce the voltage by 2.5% to avoid the possibility of inverter trips. Failure to account for this may result in lost energy production.

Additional Operating Requirements

1. The Company (DPL, ACE, Pepco) will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. This will be accomplished with a line recloser.
2. It is the Interconnection Customer's responsibility to send the data that PJM and the Company requires directly to PJM. The Interconnection Customer will grant permission for PJM to send the Company the following telemetry that the Interconnection Customer sends to PJM: real time MW, MVAR, volts, amperes, generator breaker status or inverter status, and interval MWH and MVARH.
3. The Interconnection Customer will be required to make provisions for a voice quality phone line within approximately 3 feet of each Company metering position to facilitate remote interrogation and data collection.
4. A mutually acceptable means of interrupting and disconnecting the generator with a visible break, able to be tagged and locked out, shall be worked out with Company Distribution Engineering.
5. The Company reserves the right to charge the Interconnection Customer operation and maintenance expenses to maintain the Interconnection Customer attachment facilities, including metering and telecommunications facilities, owned by the Company.
6. Study was performed with the generator on the transformer that it will be served from during normal conditions. Customer will not be allowed to generate when the feeder is served by an alternate transformer.

Summer Peak Analysis - 2020

Transmission Network Impacts

Potential transmission network impacts are as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Steady-State Voltage Requirements

None

Short Circuit

No issues identified.

Stability and Reactive Power Requirement

Not required.

Light Load Analysis - 2020

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

AC1-177

Kings Creek 138/25 kV Sub 4 MW Biomass Generator

